

Is Chain-of-Thought Reasoning of LLMs a Mirage? A Data Distribution Lens

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Chain-of-Thought (CoT) prompting has been shown to be effective in eliciting structured reasoning (i.e., CoT reasoning) from large language models (LLMs). Regardless of its popularity, recent studies expose its failures in some reasoning tasks, raising fundamental questions about the nature of CoT reasoning. In this work, we propose a data distribution lens to understand when and why CoT reasoning succeeds or fails. We hypothesize that CoT reasoning reflects a structured inductive bias learned from in-distribution data, enabling models to conditionally generate reasoning trajectories that approximate those observed during training. As such, the effectiveness of CoT reasoning is fundamentally governed by the nature and degree of distribution discrepancy between training data and test queries. Guided by this lens, we dissect CoT reasoning via three dimensions: *task*, *length*, and *format*. To test the hypothesis, we introduce DATAALCHEMY, an abstract and fully controllable environment that trains LLMs from scratch and systematically probes them under various distribution conditions. Through rigorous controlled experiments, we reveal that CoT reasoning is a brittle mirage when it is pushed beyond training distributions, emphasizing the ongoing challenge of achieving genuine and generalizable reasoning. Our code is available at GitHub: <https://github.com/ChengshuaiZhao0/DataAlchemy>.

1. Introduction

Chain-of-Thought (CoT) prompting Wei et al. (2022) has emerged as a prominent method for eliciting structured reasoning from LLMs (a.k.a., CoT reasoning). By appending a simple cue such as “Let’s think step by step”, LLMs decompose complex problems into intermediate steps, producing outputs that resemble human-like reasoning. It has been shown to be effective in tasks requiring logical inference Xu et al. (2024), mathematical problem solving Imani et al. (2023), and commonsense reasoning Wei et al. (2022). The empirical success led to CoT reasoning being seen as a promising direction towards artificial general intelligence.

However, some pioneering studies have revealed failures that challenge this optimistic view Mirzadeh et al. (2025). Stechly et al. (2024) demonstrate that LLMs fail to generalize in planning tasks, revealing a deficiency in true algorithmic reasoning. Shojaee et al. (2025) find that reasoning models experience an accuracy collapse in puzzle-solving once task complexity exceeds a critical threshold. Sun et al. (2025) demonstrate that LLMs struggle to solve complex

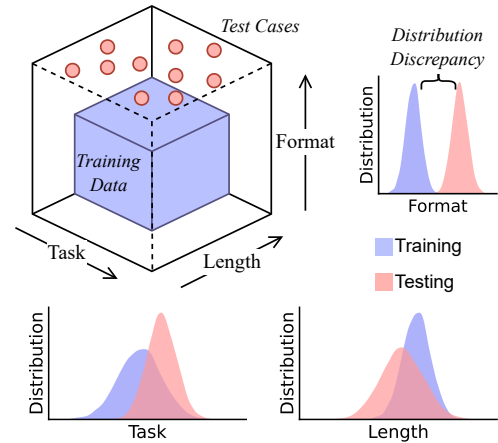


Figure 1 | The data perspective lens. CoT reasoning’s effectiveness is fundamentally bounded by the degree of distribution discrepancy introduced by *task*, *length*, and *format* between the training data and the test queries.

mathematical reasoning problem, failing to integrate or adapt learned skills to novel or creative tasks.

Considering the above *opposing* opinions, there clearly lacks an *indisputable lens to understand why and when CoT reasoning succeeds or fails*. Current evaluation approaches have intrinsic limitations that prevent them from answering the Why and When questions: (i) Narrowly defined settings. Existing frameworks focus on specific tasks and evaluate them using specific LLMs, thereby overlooking common structural patterns and characteristics. (ii) Data entanglement. Most evaluations are conducted in real-world scenarios, where the complexity precludes fully controlled experiments to isolate fine-grained factors. (iii) Data leakage. Pre-trained LLMs suffer from data leakage and benchmark containment problems, undermining the effectiveness and validity of evaluations.

In this work, we study CoT reasoning by introducing a data distribution lens. Specifically, we hypothesize that CoT reasoning reflects a structured inductive bias learned from in-distribution data, enabling models to conditionally generate reasoning trajectories that approximate those observed during training. As such, the effectiveness of CoT reasoning is fundamentally governed by the nature and degree of distribution discrepancy between training data and test queries. Guided by this lens, we revisit existing NLP tasks and identify three primary axes along which distribution shifts may occur: *task* (i.e., unseen task structures), *length* (i.e., different text lengths and reasoning lengths), and *format* (i.e., query format variant).

To tackle the issue of evaluations and validate our hypothesis, we further introduce DATAALCHEMY, an abstract, controllable, and clean environment. DATAALCHEMY provides an abstract representation system that distills various real-world NLP tasks into key components: *atoms* (i.e., token space), *elements* (i.e., text space), and *transformations* (i.e., operation space). By varying these components, we curate data that exhibits various distribution discrepancies, naturally achieving full and fine-grained control over the entire evaluation pipeline. Later, we train models *from scratch* to avoid data leakage and employ controlled experiments to rigorously test our hypotheses.

Our findings reveal that CoT reasoning works effectively when applied to (near) in-distribution data, but becomes fragile and prone to failure even under moderate distribution shifts. In some cases, LLMs generate fluent yet logically inconsistent reasoning steps. The results suggest that what appears to be structured reasoning can be a mirage, emerging from memorized or interpolated patterns in the training data rather than logical inference. Our contributions can be summarized as follows:

- ★ **Novel perspective.** We propose a *data distribution lens* for CoT reasoning, revealing that its effectiveness arises from structured inductive biases learned from in-distribution data. This lens offers a principled foundation for understanding why and when CoT reasoning succeeds or fails.
- ★ **Controllable environment.** We develop an abstract, fully controllable, and clean environment—DATAALCHEMY that abstracts NLP tasks, enabling systematic analysis of CoT reasoning under distribution discrepancies. DATAALCHEMY can serve as a research platform for probing the intrinsic behavior of LLMs, and facilitating the discovery of scientific principles.
- ★ **Rigorous investigation.** Guided by data distribution lens, we dissect the CoT reasoning via three dimensions: *task*, *length*, and *format*. Later, we curate data that reflects fine-grained factors in each dimension and conduct controlled experiments to isolate and examine each factor.
- ★ **General validity.** We train and fine-tune hundreds of LLMs with varying sizes (from 62K to 14B), architectures (e.g., GPT, LLaMA, and Qwen), and temperatures (from 1e-5 to 10). The results consistently show that the effectiveness of CoT reasoning varies with the degree of distribution discrepancy, substantiating the generality of the proposed data distribution lens.

2. Related Work

2.1. LLM Prompting and CoT

Chain-of-Thought (CoT) prompting improves large language model performance by eliciting intermediate reasoning steps for complex problems (Wei et al., 2022). Extensions include zero-shot CoT (Kojima et al., 2022), self-consistency via sampling and voting (Wang et al., 2023), and Auto-CoT, which automatically generates reasoning exemplars (Zhang et al., 2023). Beyond linear reasoning, Tree-of-Thought enables search over multiple reasoning paths (Yao et al., 2023), while SymbCoT integrates symbolic representations into CoT (Xu et al., 2024). More recent work embeds long-form CoT directly into inference, enabling reflection, error correction, and alternative reasoning strategies (Chen et al., 2025a; Guo et al., 2025; Jaech et al., 2024; Team et al., 2025; Team, 2024; Yeo et al., 2025). In this work, we investigate whether CoT reflects genuine reasoning or merely pattern interpolation.

2.2. Discussion on Illusion of LLM Reasoning

Recent work questions the robustness and faithfulness of these gains (Stechly et al., 2024). A prominent line of research shows that CoT reasoning is highly fragile: semantically irrelevant perturbations, such as distractor phrases or altered symbolic representations, can substantially degrade performance (Mirzadeh et al., 2025; Tang et al., 2023). Other studies find that models favor surface-level reasoning patterns over logical validity. Moreover, reasoning performance scales poorly with task difficulty, with models over-elaborating on simple problems and failing on harder ones (Shojaee et al., 2025). Concerns about faithfulness further arise from intervention-based analyses showing that final answers often remain unchanged when intermediate steps are corrupted or removed (Lanham et al., 2023), an effect referred to as the illusion of transparency (Bentham et al., 2024; Chen et al., 2025b). The opposing perspectives on CoT reasoning call for a systematic understanding of why and when CoT reasoning succeeds or fails.

2.3. OOD Generalization of LLMs

Out-of-distribution (OOD) generalization remains a central challenge in machine learning (Budnikov et al., 2025; Yang et al., 2024, 2023; Zhang et al., 2024). Prior work shows that pre-trained models face challenges in adapting to new settings when prompted to learn novel functions (Garg et al., 2022; Wang et al., 2024). Researchers reveal CoT prompting can partially improve OOD generalization ability, especially for tasks that require long reasoning (Shen et al., 2025; Yao et al., 2025). However, other work claims such gains are not intrinsic. For example, strong arithmetic generalization emerges only when algorithmic biases are encoded in positional representations (Cho et al., 2024), and finer-grained CoT supervision during training substantially improves OOD performance (Wang et al., 2025a). Recent studies further indicate that LLM generalizes reliably when common latent structures are shared across distributions (Li et al., 2025; Wang et al., 2025b). In light of these brilliant findings, we propose rethinking CoT reasoning through a data distribution lens, dissecting CoT reasoning into *task*, *length*, and *format*, and systematically investigating through controlled experiments. We further provide a Comparison with representative work in Appendix A.4.

3. The Proposed Data Distribution Lens

We propose the data distribution lens to understand why and when CoT reasoning succeeds or fails. We hypothesize that *CoT reasoning reflects a structured inductive bias learned from in-distribution data*,